SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

Assessment Tool Weightage: 40%

Annexure A

SHORT ANSWER TEST

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Code of Conduct:

I M.Harini (192424011) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Short Answer Test

1. A company needs to transfer a **150 MB** design file from a client computer to a remote server over a network with a bandwidth of **50 Mbps**. The Transport Layer divides the file into **1500-byte segments**, while the Data Link Layer encapsulates each segment into frames by adding **40 bytes of header and trailer**. Calculate the total number of segments and frames transmitted, the total transmission overhead introduced by the Data Link Layer, and estimate the total transmission time (ignoring propagation delay). Explain how the Transport and Data Link layers ensure reliable communication.

2. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

3. A **12,000-byte** IP packet is transmitted through a router where the Maximum Transmission Unit (MTU) is **1500 bytes**. Assuming each fragment requires a **20-byte IP header**, calculate the total number of fragments created and the total header overhead introduced. Explain how the Network Layer ensures successful delivery of fragmented packets to the destination.

4. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

5. A multimedia application transfers a **600 MB** video file. The Session Layer inserts a synchronization checkpoint after every **60 MB** of transmitted data. During transmission, a failure occurs after **390 MB** has been transferred. Determine the number of checkpoints created before failure and calculate the amount of data that must be retransmitted after recovery. Explain the importance of synchronization in the Session Layer.

6. Before transmission, the Presentation Layer compresses a **900 MB** file by **40%**. The compressed data is encrypted, adding **5%** overhead to the compressed size. Calculate the compressed file size, the encrypted file size, and the total percentage reduction compared to the original file. Explain how the Presentation Layer improves transmission efficiency and security.

7.An FTP application transfers **3 GB** of data over a network with an effective throughput of **120 Mbps**. The Application Layer divides the data into requests of **3 MB** each. Calculate the total number of requests generated and estimate the total transmission time in seconds. Discuss how the Application Layer provides reliable services to users.

8.A packet experiences the following processing delays at different OSI layers: Application Layer = **4 ms**, Transport Layer = **3 ms**, Network Layer = **5 ms**, Data Link Layer = **2 ms**, and Physical Layer = **1 ms**. The propagation delay is **18 ms**, and the transmission delay is **12 ms**. Calculate the total end-to-end delay experienced by the packet and explain how each layer contributes to the communication process.

9.A network transfers **80 MB** of useful data using frames of **1600 bytes**, where each frame contains **80 bytes** of protocol overhead added by different OSI layers. Calculate the total number of frames required, the total overhead transmitted, and the transmission efficiency as a percentage. Explain how protocol overhead affects network performance.

10. A hospital transfers a patient's medical record of **240 MB** through an IoT-enabled healthcare network. The Presentation Layer compresses the data by **30%**, the Transport Layer divides the compressed data into **1400-byte segments**, and the Data Link Layer adds **60 bytes** of overhead to each frame. Calculate the compressed file size, the total number of segments transmitted, the total protocol overhead introduced, and the final amount of data transmitted across the Physical Layer. Explain how multiple OSI layers work together to ensure secure and reliable communication.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates a thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers

Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

Question 1

Aim

To calculate the number of segments and frames transmitted for a file transfer, determine the protocol overhead introduced by the Data Link Layer, estimate the transmission time, and understand how the Transport and Data Link Layers ensure reliable communication.

Given Data

- File Size = **150 MB**
- Network Bandwidth = **50 Mbps**
- Segment Size = **1500 bytes**
- Data Link Layer Overhead = **40 bytes per frame**

Formula

File Size in Bytes

$$\text{Bytes} = \text{MB} \times 1024 \times 1024$$

Number of Segments

$$\text{Segments} = \frac{\text{Total Bytes}}{\text{Segment Size}}$$

Total Overhead

$$\text{Frames} \times \text{Overhead per Frame}$$

Transmission Time

$$\frac{\text{Total Bits}}{\text{Bandwidth}}$$

Step 1: Convert File Size

$$150 \times 1024 \times 1024$$
$$= 157{,}286{,}400 \text{ bytes}$$

Step 2: Number of Segments

Each segment contains **1500 bytes**.

$$\frac{157{,}286{,}400}{1500} = 104{,}857.6$$

Therefore,
Number of Segments = 104,858
Since each segment is placed inside one frame,

Number of Frames = 104,858

Step 3: Total Data Link Overhead

Each frame contains **40 bytes** of additional information.

$$104{,}858 \times 40 = 4{,}194{,}320 \text{ bytes}$$

≈ 4 MB

Step 4: Total Data Transmitted

$$157{,}286{,}400 + 4{,}194{,}320 = 161{,}480{,}720 \text{ bytes}$$

Convert into bits:

$$161{,}480{,}720 \times 8 = 1{,}291{,}845{,}760 \text{ bits}$$

Step 5: Transmission Time

Bandwidth

$$50 \text{ Mbps} = 50{,}000{,}000 \text{ bits/sec}$$

$$\frac{1{,}291{,}845{,}760}{50{,}000{,}000} = 25.84 \text{ seconds}$$

Final Answer

Parameter	Result
Segments	**104,858**
Frames	**104,858**
Total Overhead	**4,194,320 bytes (≈4 MB)**
Transmission Time	**25.84 seconds**

Explanation

The Transport Layer divides the large file into smaller segments so that they can be transmitted efficiently through the network. Each segment is assigned a sequence number, making it possible for the receiver to reassemble the original file in the correct order. The Transport Layer also detects lost segments and requests retransmission whenever necessary.

The Data Link Layer encapsulates every segment into a frame by adding headers and trailers. These additional bytes contain addressing information and error-detection codes such as the Cyclic Redundancy Check (CRC). If transmission errors occur, the Data Link Layer can identify corrupted frames and support reliable communication between directly connected devices.

Together, these two layers provide accurate, organized, and dependable data transmission.

Conclusion

The file is divided into **104,858 segments**, and the same number of frames are transmitted. Although the Data Link Layer introduces approximately **4 MB of protocol overhead**, it significantly improves communication reliability through framing and error detection. The complete transmission requires approximately **25.84 seconds** under the given bandwidth.

Question 2

Aim

To calculate the number of transmission windows and retransmissions using the Sliding Window Protocol and understand the importance of flow control.

Given Data

- Sliding Window Size = **6 Frames**
- Total Frames = **48**
- Frame Size = **1024 Bytes**
- One frame is lost in every window.

Formula

Number of Windows

$$\frac{\text{Total Frames}}{\text{Window Size}}$$

Retransmissions

$$\text{One Lost Frame} \times \text{Number of Windows}$$

Calculation

Step 1

$$\frac{48}{6} = 8$$

Number of transmission windows
=8

Step 2

Each window loses one frame.

$$8 \times 1 = 8$$

Retransmitted Frames
=8

Final Answer

Parameter	Result
Number of Windows	**8**
Retransmissions	**8 Frames**

Explanation

The Sliding Window Protocol allows the sender to transmit several frames before receiving acknowledgements, improving network utilization and increasing throughput. Instead of waiting for an acknowledgment after every frame, multiple frames remain in transit simultaneously.
When one frame is lost in each transmission window, only the missing frame must be retransmitted. This minimizes unnecessary retransmissions and improves communication efficiency. Flow control ensures that the sender does not overwhelm the receiver by transmitting data faster than it can process, thereby preventing buffer overflow and packet loss.

Conclusion

Using a window size of six frames, the sender requires **8 transmission windows** to send all **48 frames**. Since one frame is lost in every window, a total of **8 retransmissions** are required. Flow control increases efficiency by balancing the transmission rate between sender and receiver.

Question 3

Aim

To determine the number of fragments generated when an IP packet exceeds the Maximum Transmission Unit (MTU) and calculate the resulting IP header overhead.

Given Data
- IP Packet Size = **12,000 Bytes**
- MTU = **1500 Bytes**
- IP Header = **20 Bytes**

Formula

Payload per Fragment

$$1500 - 20$$

Fragments

$$\frac{\text{Packet Size}}{\text{Payload}}$$

Header Overhead

$$\text{Fragments} \times 20$$

Step 1

Payload available in each fragment

$$1500 - 20 = 1480 \text{ bytes}$$

Step 2

$$\frac{12000}{1480} = 8.1$$

Therefore,

Number of Fragments = 9

Step 3

Total Header Overhead

$$9 \times 20 = 180 \text{ bytes}$$

Final Answer

Parameter	Result
Fragments Created	**9**
Header Overhead	**180 Bytes**

Explanation

When a packet is larger than the MTU supported by a network, the Network Layer divides it into smaller fragments. Every fragment receives its own IP header containing identification, fragment offset, and control information. At the receiving end, the Network Layer uses these values to correctly reassemble the original packet.
Although fragmentation increases protocol overhead because each fragment requires an additional header, it enables large packets to travel across networks with different MTU sizes without data loss.

Conclusion

A **12,000-byte IP packet** is divided into **9 fragments**, resulting in an additional **180 bytes of IP header overhead**. Fragmentation ensures successful packet delivery across networks with different MTU limitations while maintaining reliable communication.

Question 4

Aim

To calculate the number of transmission windows and retransmissions using the Sliding Window Protocol and explain how flow control improves communication efficiency.

Introduction

The **Sliding Window Protocol** is a Transport Layer mechanism used to improve communication efficiency. Instead of sending one frame and waiting for its acknowledgment, the sender can transmit multiple frames before receiving acknowledgments. This reduces waiting time, improves bandwidth utilization, and increases network throughput.
Flow control ensures that the sender transmits data at a rate that the receiver can handle, preventing data loss due to receiver buffer overflow.

Given Data

- Sliding Window Size = **6 Frames**
- Total Frames = **48**
- User Data per Frame = **1024 Bytes**
- One frame is lost in every transmission window.

Formula

Number of Windows

$$\text{Number of Windows} = \frac{\text{Total Frames}}{\text{Window Size}}$$

Number of Retransmissions

$$\text{Retransmissions} = \text{Number of Windows} \times 1$$

Calculation

Step 1: Calculate Number of Windows

$$\frac{48}{6} = 8$$

Therefore,
Number of Windows = 8

Step 2: Calculate Retransmissions

One frame is lost in each transmission window.

$$8 \times 1 = 8$$

Therefore,
Retransmitted Frames = 8

Final Answer

Parameter	Value
Total Frames	48
Window Size	6
Number of Windows	**8**
Retransmissions	**8 Frames**

Explanation

The Sliding Window Protocol allows multiple frames to be transmitted without waiting for acknowledgments after every frame. This greatly improves the utilization of available bandwidth.
Whenever a frame is lost, only that particular frame is retransmitted instead of sending all frames again. This reduces network traffic and saves transmission time.
Flow control ensures that the sender does not send data faster than the receiver can process it. Without flow control, receiver buffers may overflow, resulting in packet loss and retransmissions.
This mechanism is widely used in TCP to provide reliable communication over the Internet.

Conclusion

The sender requires **8 transmission windows** to transmit **48 frames**. Since one frame is lost in every window, **8 retransmissions** are needed. Sliding Window Protocol significantly improves communication speed while maintaining reliable data delivery.

Question 5

Aim

To determine the number of synchronization checkpoints created during file transfer and calculate the amount of data that must be retransmitted after a communication failure.

Introduction

The **Session Layer** manages communication sessions between devices. One of its important functions is synchronization. During long file transfers, the Session Layer inserts synchronization checkpoints at regular intervals.
If transmission fails, communication resumes from the last checkpoint rather than starting again from the beginning. This saves bandwidth, reduces transmission time, and improves communication reliability.

Given Data

- Video File Size = **600 MB**
- Synchronization Checkpoint = Every **60 MB**
- Failure Occurs After **390 MB**

Formula

Number of Checkpoints

$$\text{Checkpoints} = \frac{\text{Data Successfully Passed}}{\text{Checkpoint Interval}}$$

Data to Retransmit

$$\text{Failure Point} - \text{Last Checkpoint}$$

Calculation

Step 1: Determine Checkpoints
Synchronization occurs every **60 MB**
Checkpoints are created at

- 60 MB
- 120 MB
- 180 MB
- 240 MB
- 300 MB
- 360 MB

The next checkpoint would be at **420 MB**, but failure occurs before reaching it.
Therefore,
Number of Checkpoints = 6

Step 2: Calculate Retransmitted Data

Failure occurs at
390 MB
Last checkpoint
360 MB
Retransmitted data

$$390 - 360 = 30 \text{ MB}$$

Final Answer

Parameter	**Result**
Checkpoints Created	**6**
Data to Retransmit	**30 MB**

Explanation

The Session Layer inserts synchronization points at regular intervals during communication.
These checkpoints divide large transfers into manageable sections.
When a communication failure occurs after **390 MB**, the receiver resumes transmission from the last completed checkpoint at **360 MB**.
Only the remaining **30 MB** must be retransmitted instead of the entire **390 MB**, greatly reducing transmission time and network traffic.

Synchronization is particularly important in multimedia streaming, cloud storage, banking systems, and large database transfers where restarting an entire transmission would be inefficient.

Conclusion

The Session Layer creates **6 synchronization checkpoints** before failure occurs. Only **30 MB** of data needs to be retransmitted after recovery, demonstrating how synchronization improves network efficiency and reliability.

Question 6

Aim
To calculate the compressed and encrypted file sizes after Presentation Layer processing and determine the overall reduction in transmitted data.

Introduction

The **Presentation Layer** prepares data before transmission by performing functions such as:

- Data Compression
- Data Encryption
- Data Formatting
- Character Encoding

Compression reduces file size, resulting in faster transmission and lower bandwidth usage. Encryption protects data from unauthorized access while it travels through the network.

Given Data

- Original File Size = **900 MB**
- Compression = **40%**
- Encryption Overhead = **5%**

Formula

Compressed File Size

$$\text{Compressed Size} = \text{Original Size} \times (100 - 40)\%$$

Encrypted Size

$$\text{Encrypted Size} = \text{Compressed Size} \times 105\%$$

Percentage Reduction

$$\frac{\text{Original} - \text{Encrypted}}{\text{Original}} \times 100$$

Calculation

Step 1: Compression
Remaining data

$$100 - 40 = 60\%$$

Compressed file

$$900 \times 0.60 = 540 \text{ MB}$$

Step 2: Encryption

Encryption adds **5%**

$$540 \times 1.05 = 567 \text{ MB}$$

Step 3: Percentage Reduction

Original
900 MB
Final
567 MB

$$\frac{900 - 567}{900} \times 100 = 37\%$$

Final Answer

Parameter	**Result**
Original File	900 MB
Compressed File	**540 MB**
Encrypted File	**567 MB**
Overall Reduction	**37%**

Explanation

The Presentation Layer first compresses the data, reducing the file size by **40%**. Smaller files require less bandwidth and can be transmitted more quickly across the network.
After compression, encryption is applied to ensure confidentiality and prevent unauthorized access. Although encryption adds **5%** overhead, the final encrypted file remains significantly smaller than the original.
The Presentation Layer therefore improves both communication efficiency and data security.
This layer is commonly used in secure Internet applications such as HTTPS, online banking, cloud storage, video conferencing, and secure email systems.

Conclusion

The original **900 MB** file is compressed to **540 MB** and becomes **567 MB** after encryption. Even after adding encryption overhead, the total transmitted data is **37% smaller** than the original file, demonstrating how the Presentation Layer enhances both efficiency and security.

Question 7

Aim

To calculate the total number of FTP requests generated during a file transfer, estimate the transmission time, and explain how the Application Layer provides reliable services to users.

Introduction

The **Application Layer** is the topmost layer of the OSI model. It provides network services directly to users and applications, such as file transfer, email, web browsing, and remote login. The **File Transfer Protocol (FTP)** is an Application Layer protocol used to transfer files between a client and a server over a network.
In this problem, a 3 GB file is transferred using FTP, and the data is divided into requests of 3 MB each.

Given Data
- Total Data = **3 GB**
- Effective Throughput = **120 Mbps**
- Request Size = **3 MB**

Formula

Convert GB to MB

$$\text{Data (MB)} = \text{Data (GB)} \times 1024$$

Number of Requests

$$\text{Number of Requests} = \frac{\text{Total Data (MB)}}{\text{Request Size (MB)}}$$

Transmission Time

$$\text{Transmission Time} = \frac{\text{Total Data (bits)}}{\text{Bandwidth (bps)}}$$

Calculation

Step 1: Convert Data into MB

$$3 \times 1024 = 3072 \text{ MB}$$

Step 2: Calculate Number of Requests

Each request transfers **3 MB**.

$$\frac{3072}{3} = 1024$$

Therefore,

Number of Requests = 1024

Step 3: Convert Data into Bits

$$3072 \times 1024 \times 1024 = 3{,}221{,}225{,}472 \text{ bytes}$$

Convert bytes to bits:

$$3{,}221{,}225{,}472 \times 8 = 25{,}769{,}803{,}776 \text{ bits}$$

Step 4: Calculate Transmission Time

Bandwidth:

$$120 \text{ Mbps} = 120{,}000{,}000 \text{ bits/sec}$$

$$\frac{25{,}769{,}803{,}776}{120{,}000{,}000} = 214.75 \text{ seconds}$$

Approximately,
Transmission Time ≈ 215 seconds (3.58 minutes)

Final Answer

Parameter	Result
Total Data	**3 GB (3072 MB)**
Request Size	**3 MB**
Number of Requests	**1024**
Transmission Time	**214.75 seconds (≈ 3.58 minutes)**

Explanation

The **Application Layer** provides network services directly to end users. In this scenario, FTP divides the 3 GB file into **1024 requests**, with each request carrying **3 MB** of data. The effective throughput of **120 Mbps** determines how quickly the data can be transferred over the network.
The Application Layer ensures reliable file transfer by establishing a connection between the client and server, authenticating users, managing file transfer requests, and reporting transfer status. It also supports error handling and allows interrupted transfers to be restarted in many implementations, improving reliability and user experience.

Conclusion

A **3 GB** file transferred using FTP generates **1024 requests** of **3 MB** each. With an effective throughput of **120 Mbps**, the estimated transmission time is **214.75 seconds (approximately 3.58 minutes)**. The Application Layer ensures efficient and reliable file transfer by managing communication between the user application and the network.

Question 8

Aim

To calculate the total end-to-end delay experienced by a packet and explain the contribution of each OSI layer during communication.

Given Data

- Application Layer Delay = **4 ms**
- Transport Layer Delay = **3 ms**
- Network Layer Delay = **5 ms**
- Data Link Layer Delay = **2 ms**
- Physical Layer Delay = **1 ms**
- Propagation Delay = **18 ms**
- Transmission Delay = **12 ms**

Formula

$$\text{Total End-to-End Delay} = \text{Processing Delay} + \text{Propagation Delay} + \text{Transmission Delay}$$

Calculation

Processing Delay:

$$4 + 3 + 5 + 2 + 1 = 15 \text{ ms}$$

Total Delay:

$$15 + 18 + 12 = 45 \text{ ms}$$

Final Answer

Parameter	**Value**
Total Processing Delay	**15 ms**
Propagation Delay	**18 ms**
Transmission Delay	**12 ms**
Total End-to-End Delay	**45 ms**

Explanation

The total delay experienced by a packet is the sum of all processing delays at different OSI layers, the propagation delay, and the transmission delay.

- The **Application Layer** prepares user data for transmission.
- The **Transport Layer** performs segmentation, flow control, and error recovery.
- The **Network Layer** determines the best routing path.
- The **Data Link Layer** frames the data and performs error detection.
- The **Physical Layer** converts the frame into electrical or optical signals for transmission.
- **Propagation delay** is the time required for the signal to travel through the transmission medium.

- **Transmission delay** is the time taken to place all bits onto the communication channel.

The combined delay is **45 ms**, which represents the total time required for the packet to reach the destination.

Conclusion

The packet experiences a total end-to-end delay of **45 ms**. Each OSI layer contributes a small processing delay, while propagation and transmission delays account for the movement of the packet across the network.

Question 9

Aim

To calculate the number of frames required, the total protocol overhead transmitted, and the transmission efficiency of a network.

Given Data

- Useful Data = **80 MB**
- Frame Size = **1600 bytes**
- Protocol Overhead = **80 bytes per frame**

Formula

Useful Payload per Frame

$$1600 - 80 = 1520 \text{ bytes}$$

Number of Frames

$$\frac{\text{Useful Data}}{\text{Payload per Frame}}$$

Protocol Overhead

$$\text{Frames} \times \text{Overhead}$$

Transmission Efficiency

$$\frac{\text{Payload}}{\text{Frame Size}} \times 100$$

Calculation

Convert useful data into bytes:

$$80 \times 1024 \times 1024 = 83{,}886{,}080 \text{ bytes}$$

Number of Frames:

$$\frac{83{,}886{,}080}{1520} = 55{,}188.21$$

Rounded up,
Frames = 55,189
Protocol Overhead:

$$55{,}189 \times 80 = 4{,}415{,}120 \text{ bytes}$$

Transmission Efficiency:

$$\frac{1520}{1600} \times 100 = 95\%$$

Final Answer

Parameter	Result
Frames Required	**55,189**
Protocol Overhead	**4,415,120 bytes (≈4.21 MB)**
Transmission Efficiency	**95%**

Explanation

Every transmitted frame contains both user data and protocol information. The protocol overhead includes headers and trailers that provide addressing, error detection, synchronization, and flow control.
Although the overhead consumes additional bandwidth, it is necessary for reliable communication. In this problem, each frame carries **1520 bytes of useful data** and **80 bytes of protocol information**, resulting in a transmission efficiency of **95%**.
Higher protocol overhead reduces network efficiency because more bandwidth is spent transmitting control information instead of useful data. Therefore, network protocols are designed to minimize overhead while maintaining reliability and security.

Conclusion

The network requires **55,189 frames** to transfer the data. A total protocol overhead of **4,415,120 bytes** is transmitted, and the network achieves a high transmission efficiency of **95%**.

Question 10

Aim

To calculate the compressed file size, the total number of segments, protocol overhead, and the final amount of data transmitted through the Physical Layer in an IoT-enabled healthcare network.

Given Data

- Original Medical Record = **240 MB**
- Compression = **30%**
- Segment Size = **1400 bytes**
- Data Link Layer Overhead = **60 bytes per frame**

Formula

Compressed File Size

$$\text{Compressed Size} = \text{Original Size} \times (100 - 30)\%$$

Number of Segments

$$\frac{\text{Compressed Bytes}}{\text{Segment Size}}$$

Protocol Overhead

$$\text{Segments} \times 60$$

Final Data Transmitted

$$\text{Compressed Data} + \text{Protocol Overhead}$$

Calculation

Step 1: Compressed File Size

$$240 \times 0.70 = 168 \text{ MB}$$

Convert to bytes:

$$168 \times 1024 \times 1024 = 176{,}160{,}768 \text{ bytes}$$

Step 2: Number of Segments

$$\frac{176{,}160{,}768}{1400} = 125{,}829.12$$

Rounded up,
Segments = 125,830

Step 3: Protocol Overhead

$$125{,}830 \times 60 = 7{,}549{,}800 \text{ bytes}$$

≈ **7.2 MB**

Step 4: Final Data Transmitted

$$176{,}160{,}768 + 7{,}549{,}800 = 183{,}710{,}568 \text{ bytes}$$

≈ **175.2 MB**

Final Answer

Parameter	Result
Compressed File Size	**168 MB**

Parameter	Result
Number of Segments	**125,830**
Protocol Overhead	**7,549,800 bytes (≈7.2 MB)**
Final Data Transmitted	**183,710,568 bytes (≈175.2 MB)**

Explanation

In this healthcare network, multiple OSI layers work together to ensure efficient, reliable, and secure communication.

- The **Presentation Layer** compresses the medical record by **30%**, reducing bandwidth usage and enabling faster transmission.
- The **Transport Layer** divides the compressed file into **1400-byte segments**, ensuring reliable end-to-end delivery through sequencing, acknowledgments, and error recovery.
- The **Data Link Layer** encapsulates each segment into a frame and adds **60 bytes** of header and trailer information for addressing and error detection.
- Finally, the **Physical Layer** transmits the complete bit stream over the communication medium.

This layered approach ensures that sensitive patient information is transferred accurately, efficiently, and reliably across the IoT-enabled healthcare network.

Conclusion

The **240 MB** medical record is compressed to **168 MB**, divided into **125,830 segments**, and incurs **7,549,800 bytes** of protocol overhead. The total data transmitted through the Physical Layer is **183,710,568 bytes (approximately 175.2 MB)**. These calculations demonstrate how different OSI layers cooperate to provide efficient, secure, and reliable communication.